BRIDGE_C_35 order5

order5 REPORT (descriptive notes; verdicts as in RESULTS.md)

PREREG sha256 5c40ee5b…fe49 (unchanged since hashing at about 2:10 PM CDT). Script hashes: scripts/SHA256SUMS. Process deviations are in NOTES.md. All verdicts below are the preregistered ones.

Category A (exact or derived; proof sketches flagged)

- **Cone structure (finite algebra).** w(v) = ½(sgn B(c1,v) − sgn B(c2,v)) with c1 = (−1,1), c2 = (−4,1); Q(c1) = −5/2, Q(c2) = −17/2, B(c1,c2) = −20. The wall directions c1^⊥ = (0,1) (Q = 7/2) and c2^⊥ = (21,−1) (Q = 595/2) are the order3 labels. The orthant's odd part is the Zwegers pair (c1, c′ = (−7,2)), with Q(c′) = −35 and c′^⊥ = (1,0) (Q = 1), plus the axis false thetas F. The double orthant (the even part) is not of Zwegers type.
- **v-signs versus ν-signs (finite algebra).** H_ν − H = U_H and ρ_O(ν)-series − ½(sgn n + sgn k)-series = U_O. Both are finite sums of full unary theta lines (PREREG §1.2).
- **Completion constants (derived; Zwegers' theorem cited).** The completion is Ĥ = Σ_ν ½[E(√10·ν1·√y) + E((ν1+21ν2)·√(2y/17))]·e(Q(ν)τ + B(ν,β)), and for the orthant pair the term E((ν1+21ν2)·√(2y/17)) is replaced by E(√35·ν2·√y). Here E(z) = erf(√π z), and at the radial point y = t/(2π).
- **Shadow (calculus).** The shadow is *$\partial_{\bar\tau}\hat H = -(i/(2\sqrt{y}))\,\Sigma_j\,\sigma_j\,\sqrt{|Q(c_j)|}\,\Sigma_\nu\, w_j\, e(Q(u_j)s_j^2\tau + B(\nu,\beta))\cdot \mathrm{conj}(e(|Q(c_j)|w_j^2\tau))$. Its weight-3/2 parts are the unary thetas $\Sigma\, n\, e(5n^2\tau/2)$ with $n \in \alpha 1+Z$ (c1, unimodular split, paired with $\theta\{7/2\}$)* and Σ R e(R²τ/34) with R ∈ α1+21α2+Z (c2, glued over 17 classes with θ_{35/34}, i.e. the wall Q = 595/2).
- **Radial consequences, proof sketch only (PREREG §1.4).** Ĥ has no t^{−1/2} and no t^0 term. A2(H) ≡ 0. A2(S_odd) = A2(F). The E-argument constants do not enter A1 or A2. **These are listed as proof sketches, not as completed proofs. Their status rests on the Category B checks P1 and P2 below.**

Category B (numerical)

Task (a)

test	statistic	threshold	verdict
Z controls: holomorphic H_v; E-args × 1.25	r = 0.300; 0.109 (Ĥ), 0.112 (Ĝ_O)	must be > 1e−4	**controls fail, as required**
Z: Ĥ obeys the fixed 35-component law vector S-law, weight 1, Ĥ(−1/τ) = (τ/√35)·M·Ĥ(τ), 120 interior τ	r = 6.0e−16 (truncation 1.1e−16)	≤ 1e−10	**pass**
Z: Ĝ_O (orthant odd pair)	r = 6.1e−16	≤ 1e−10	**pass**
P3: finite-difference ∂_τĤ vs the shadow formula (24 cases)	max rel. err 4.5e−9	≤ 1e−6	**pass**
P1: max |A2(H)| over 105 × 216 × {x, −1/x} (mp, dps 30)	1.1e−30 (|A2| up to 18.4)	≤ 1e−11	**pass**
P2: max |A2(S_odd) − A2(F)|	6.8e−15	≤ 1e−11	**pass**
check: A2 of the full lines U_H, U_O	4.4e−30	(should be 0)	consistent
sign control (A0, XS216)	rank 36, closed at w=1; joint 72 at w=−1; fixed-matrix residual 7.7e−16	gating	**passed**

Completed-law tests on XS216 (SVD threshold 1e−9). The fixed-matrix residual is not defined for the odd combinations (NOTES item 4). | object | rank | joint (w) | fixed-matrix residual | closed | |—|—|—|—|—| | A1 (reference) | 105 | 210 (½), 210 (1) | 1.82, 4.79 | no | | A1^c = A1(orth) + A1(U_H) | 105 | 210 (½), 210 (1) | 1.53, 4.27 | **no** | | A1^c_odd = A1(−U_O + 2U_H + F) | 103 | 206 (½), 206 (1) | – | **no** | | A2^c = A2(orth) (= A2, by P1) | 73 | 146 (0), 146 (½) | 0.69, 0.82 | **no** | | A2^c_odd = A2(F) | 99 | 190 (0), 190 (½) | – | **no** |

All four “not closed” predictions hold.

Shadow verdict (preregistered rule): - The shadow of the cone part H is located (Category B confirmation of the Category A derivation). Z passes with both controls failing, P1 passes, and P3 passes. It is the pair of weight-3/2 unary thetas of §1.5, attached to the walls Q = 7/2 and 595/2. - **The shadow of the full C is not located.** The

completed A1^c and A2^c do not satisfy the fixed S-law. The remaining obstruction sits in the orthant piece: the axis false thetas F (Q = 1, 7/2) and the uncompleted double-orthant part.

Task (b): sharpened A2 mixed law

- Points XSb: 359.
- B = [A2, |x|^{½}A1, |x|A0]: numerical rank 214 at dps 30, with a gap of 1.4e27 at the cut (last pivot 7.6e−3, first rejected 5.4e−30). Rows/unknowns = 1.68, which meets the ≥ 1.5 requirement.
- Positive control $|x|^2$A0(−1/x): ε = 2.4e−31 (≤ 1e−20, passes). Negative control A0(−1/x): ε ≥ 0.53 (passes). **Valid.**
- **ε = max residual of A2(−1/x) = 0.389 (min 0.276) ≥ 1e−6 → mixed law FALSIFIED (Category B), decisively.**
- Cross-check of the mp entries against float lib3c: see the bcross line below. t-grid anchors (brute force, t_j = 2^{−j}·0.5/(297.5(2m)²)): x = 1/8 gives err 1.5e−7 and x = 3/8 gives err 2.5e−7. **Pass, 2/2.**
- bcross: max relative difference between the mp entries and float lib3c over all 359 × 105 × 6 entries = 3.7e−13 (≤ 1e−12, **pass**).

Status summary

- **Category A, unchanged:** the bridge through order t^0. A1 = D + R + W[1] + W[7/2] + W[595/2].
- **Category A, new:** the identities of PREREG §1.1–1.2 and the shadow formula §1.5. Ĥ's fixed S-law, by citation of Zwegers' theorem.
- **Proof sketches, not upgraded to Category A:** A2(H) ≡ 0 and A2(S_odd) = A2(F).
- **Category B:** everything numerical above.

Appendix: order5 notes (descriptive; disclosures)

1. **zwegers.py, first launch.** The first launch (about 2:14 PM CDT) hit float overflow (inf·0 → NaN) and was killed. No statistic was read from it. The fix was a rewrite of the series summation in an overflow-safe form (erfcx, masked exponent). Tests, thresholds, τ-grid and controls were unchanged. SHA256SUMS was regenerated before the rerun.
2. **bchunk.py, first launch.** The first launch failed on a NameError (`_ET` is not exported by `import *`), before any data was written. It was fixed and relaunched.
3. **Order deviation.** The mp-vs-float cross-check (bcross.py) ran after btest.py, not before. It passed at 3.7e−13, so the verdict is unaffected. Recorded verdicts stand.
4. **Odd combinations.** S_{α,β} depends on the representative α, not only on α mod Z^2, because the weight wt is applied to ν, not to ν. So S_{−α,−β} with $-\alpha \notin [0,1)^2$ is not a member of the 105-family up to a phase. Two consequences:
   - the fixed-matrix residual is undefined for the odd objects, and only the rank test was used, as scripted;
   - the odd objects can have a rank above 73 (A2^c_odd has rank 99). The completion Ĥ uses ν-signs and depends only on α mod Z^2, so test Z is unaffected.
5. **P2 precision.** The P2 residual is 6.8e−15, not 1e−30 like P1. This is consistent with double rounding of three O(10) numbers after conversion to complex. It is far below

the 1e−11 pass line.

6. **Reading (descriptive, Category B):**
   - Because A2(H) ≡ 0, the whole O(1) coefficient A2 comes from the closed-orthant piece, so A2^c = A2.
   - The H part contributes to A1 only through the full-line unary thetas U_H and the jump-sawtooth radial limit of the Eichler integral of its shadow.
   - The erf constants (√10, √(2/17), √35) are invisible in A1 and A2. They are fixed only by the S-law (test Z), where a 25% change gives residual 0.11.
7. **Rank of Ĥ.** Ĥ on 120 interior τ has rank 44 (Ĝ_O also 44). This is descriptive.
8. **Still open:** completion of the double orthant and of the axis false thetas; smoothness of their S-defects; a written full proof of §1.4.
9. **Process.** nice, OMP_NUM_THREADS=1, ulimit -v 3000000. No push, no external sends. Nothing outside order5/ was modified.